SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATINO SYSTEMS

WINTER SEMERSTER 2025-26

FINAL ASSESSMEET TEST

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SET - 3

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COURSE : Agile Development process and Devops Lab

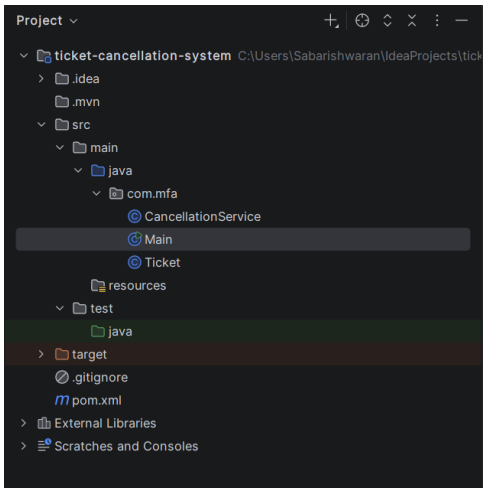
FACULTY : Dr. N . BALA GANESH

Online Ticket Canellation system

Problem Statement :

Ensure tickets are cancelled and refunded correctly

Project Structure:



Ticket Class Implementation

Main.java | pom.xml (ticket-cancellation-system) | Ticket.java | CancellationService.java

```java
package com.mfa;

public class Ticket {  no usages
    private int ticketId;  1 usage
    private double price;  2 usages
    private int hoursBeforeTravel;  2 usages

    public Ticket(int ticketId, double price, int hoursBeforeTravel) {  no usages
        this.ticketId = ticketId;
        this.price = price;
        this.hoursBeforeTravel = hoursBeforeTravel;
    }

    public double getPrice() {  no usages
        return price;
    }

    public int getHoursBeforeTravel() {  no usages
        return hoursBeforeTravel;
    }
}
```

Refund Calculation Logic

Main.java | Ticket.java | CancellationService.java

```java
package com.mfa;

public class CancellationService {  no usages

    public double calculateRefund(Ticket ticket) {  no usages
        int hours = ticket.getHoursBeforeTravel();
        double price = ticket.getPrice();

        if (hours > 48) {
            return price * 0.9;
        } else if (hours >= 24) {
            return price * 0.5;
        } else {
            return 0;
        }
    }
}
```

Main Class for Execution

```java
package com.mfa;

public class Main {
    public static void main(String[] args) {

        Ticket ticket = new Ticket( ticketId: 101,  price: 1000,  hoursBeforeTravel: 30);
        CancellationService service = new CancellationService();

        double refund = service.calculateRefund(ticket);

        System.out.println("Ticket Cancelled!");
        System.out.println("Refund Amount: ₹" + refund);
    }
}
```

Execution Output Showing Refund Calculation

```
Run  Main
"C:\Program Files\Java\jdk-21.0.10\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1\lib\idea_rt.jar=58683" -Dfile.encoding=UTF-8 -Dsun.stdout
Ticket Cancelled!
Refund Amount: ₹500.0

Process finished with exit code 0
```

JUnit Test Cases for Refund Logic

```java
package com.mfa;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class CancellationServiceTest {

    CancellationService service = new CancellationService();  3 usages

    @Test
    public void testFullRefund() {
        Ticket t = new Ticket( ticketId: 1,  price: 1000,  hoursBeforeTravel: 60);
        assertEquals( expected: 900, service.calculateRefund(t));
    }

    @Test
    public void testHalfRefund() {
        Ticket t = new Ticket( ticketId: 2,  price: 1000,  hoursBeforeTravel: 30);
        assertEquals( expected: 500, service.calculateRefund(t));
    }

    @Test
    public void testNoRefund() {
        Ticket t = new Ticket( ticketId: 3,  price: 1000,  hoursBeforeTravel: 10);
        assertEquals( expected: 0, service.calculateRefund(t));
    }
}
```

Successful Execution of JUnit Test Cases

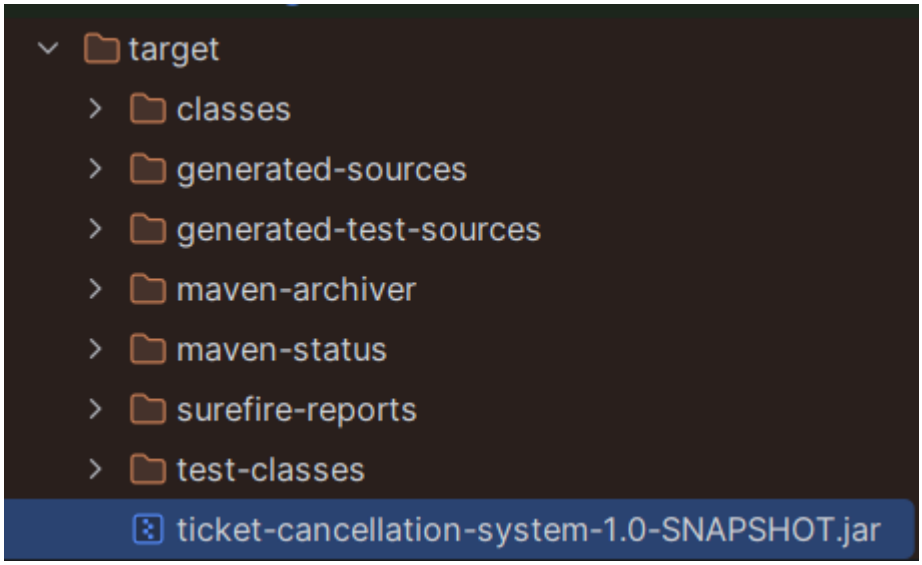
```
Run   CancellationServiceTest
CancellationServiceTest (cc 95 ms      3 tests passed  3 tests total, 95 ms
    testFullRefund()        88 ms      "C:\Program Files\Java\jdk-21.0.10\bin\java.exe" ...
    testHalfRefund()         4 ms
    testNoRefund()           3 ms      Process finished with exit code 0
```

Maven Build Successful with Test Execution

mvn clean install

```
Terminal   Local
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.mfa.CancellationServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.082 s - in com.mfa.CancellationServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ ticket-cancellation-system ---
[INFO] Building jar: C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system\target\ticket-cancellation-system-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ ticket-cancellation-system ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom (2.2 kB at 12 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom (23 kB at 150 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom (2.2 kB at 8.2 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar (196 kB at 81 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (157 kB at 123 kB/s)
[INFO] Installing C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system\pom.xml to C:\Users\Sabarishwaran\.m2\repository\com\mfa\ticket-cancellation-system\1.0-SNAPS
HOT\ticket-cancellation-system-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system\target\ticket-cancellation-system-1.0-SNAPSHOT.jar to C:\Users\Sabarishwaran\.m2\repository\c
om\mfa\ticket-cancellation-system\1.0-SNAPSHOT\ticket-cancellation-system-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  36.831 s
[INFO] Finished at: 2026-04-16T05:58:39+05:30
[INFO] ------------------------------------------------------------------------

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system>
```

Generated JAR File after Maven Build



Connect IntelliJ Project to Git

git init

git add .

git commit -m "Initial commit - Ticket Cancellation System"

```
Terminal   Local

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> git init
Initialized empty Git repository in C:/Users/Sabarishwaran/IdeaProjects/ticket-cancellation-system/.git/

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/mfa/Main.java', LF will be replaced by CRLF the next time Git touches it

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> git commit -m "Initial commit - Ticket Cancellation System"
[master (root-commit) e83f43d] Initial commit - Ticket Cancellation System
 9 files changed, 181 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/encodings.xml
 create mode 100644 .idea/misc.xml
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/mfa/CancellationService.java
 create mode 100644 src/main/java/com/mfa/Main.java
 create mode 100644 src/main/java/com/mfa/Ticket.java
 create mode 100644 src/test/java/com/mfa/CancellationServiceTest.java

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system>
```

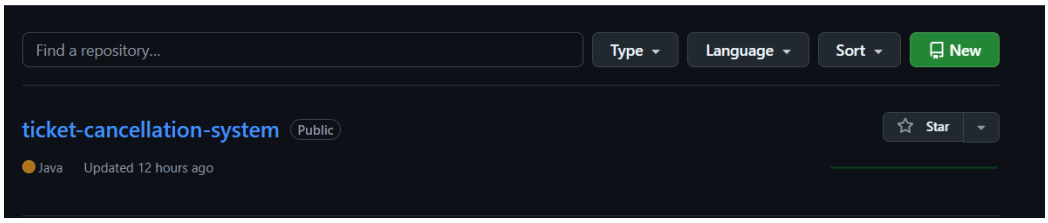
Link GitHub Repo

git remote add origin https://github.com/sabarishwaran2023-source/ticket-cancellation-system.git

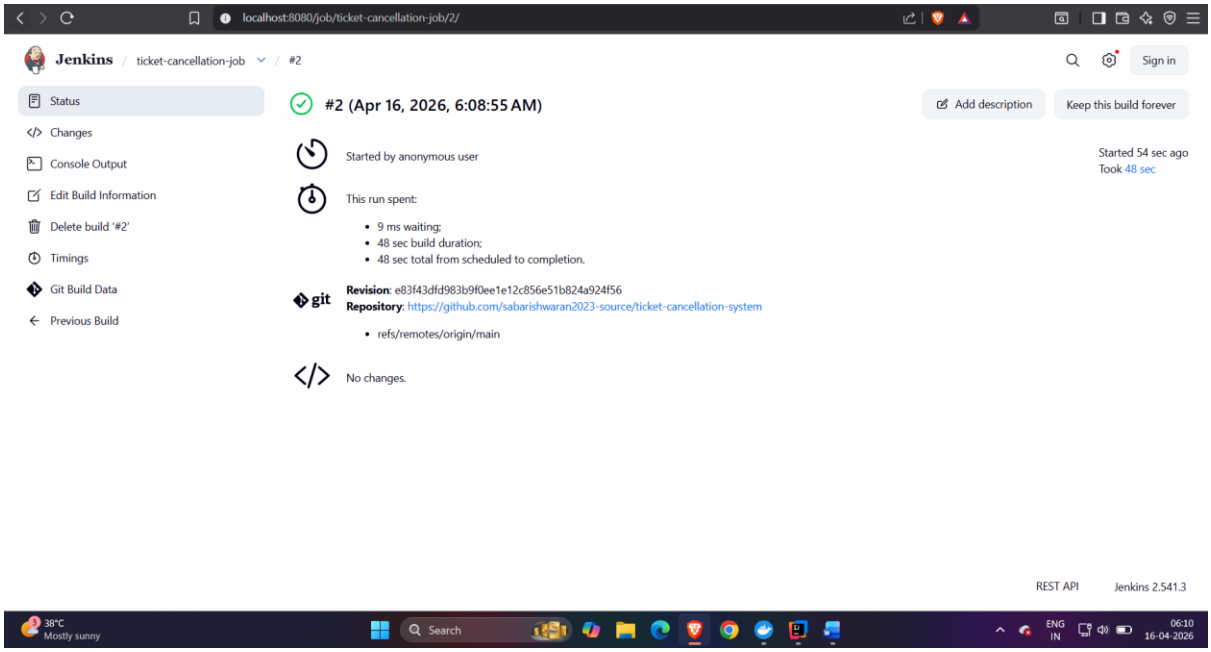
git branch -M main

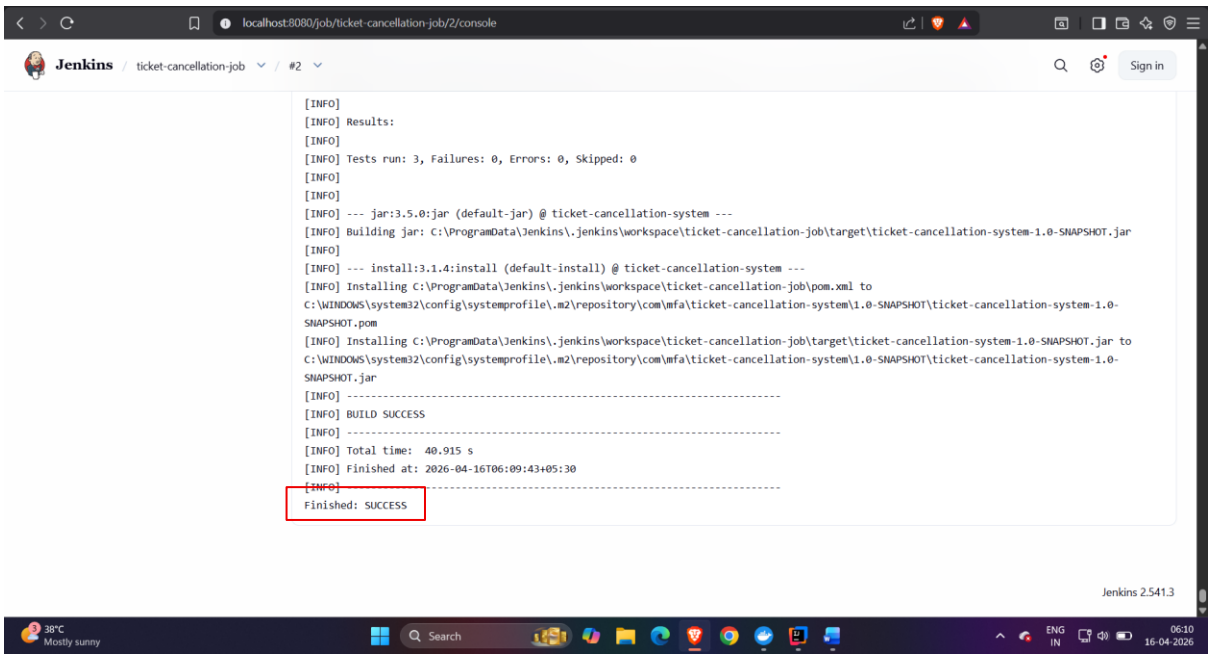
git push -u origin main

```
Terminal   Local

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> git remote add origin https://github.com/sabarishwaran2023-source/ticket-cancellation-system

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> git branch -M main

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> git push -u origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 16 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (21/21), 3.02 KiB | 238.00 KiB/s, done.
Total 21 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sabarishwaran2023-source/ticket-cancellation-system
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



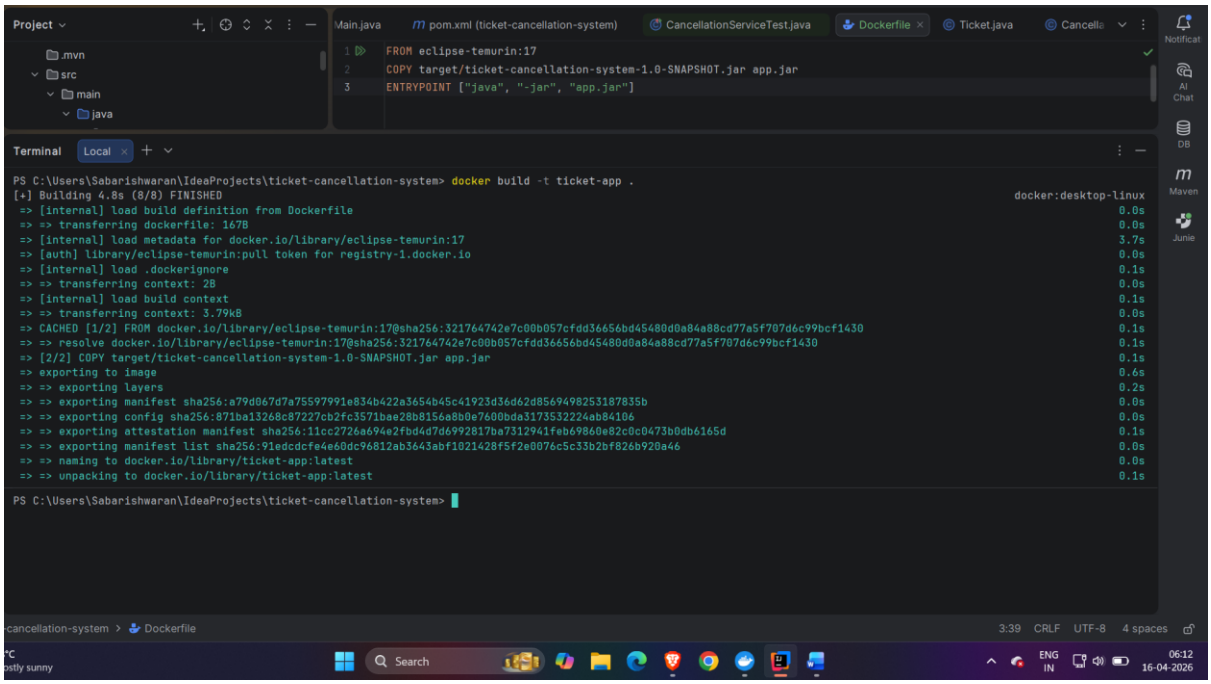
Successful Jenkins Build Execution



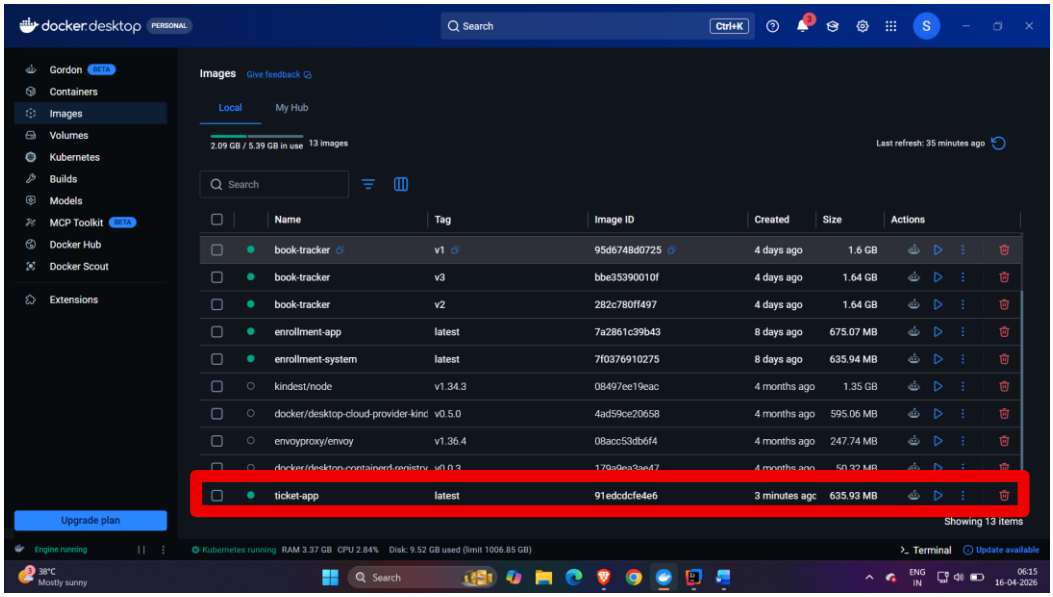


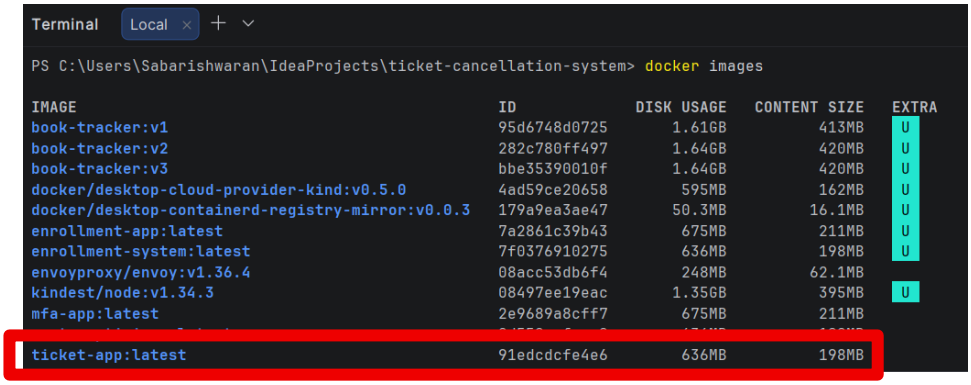
Docker file and its Output :

docker build -t ticket-app .



Docker Image Created Successfully (docker images)





Successful Execution of Application inside Docker Container

docker run ticket-app

```
PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> docker build -t ticket-app .
[+] Building 3.0s (7/7) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 167B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 3.88kB
 => CACHED [1/2] FROM docker.io/library/eclipse-temurin:17@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430
 => => resolve docker.io/library/eclipse-temurin:17@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430
 => [2/2] COPY target/ticket-cancellation-system-1.0-SNAPSHOT.jar app.jar
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:14a94cdd971835a939d84e2da3f0dc2e11cd4a83c95f5d42d1beeec3ca025c24
 => => exporting config sha256:eedf425d81c76639553b6c4a2fd625d397a95d0f99e9b22be9ef6419f2f8f9df
 => => exporting attestation manifest sha256:11da8bbbf0acabb2f361b259ff608054f942dd4fa86530d683b9e7edb1e01ed8
 => => exporting manifest list sha256:ff61e4c21f7a72ecc8f838a5a1cccd418da56ac63be0b94c882ce03bffee5b5f
 => => naming to docker.io/library/ticket-app:latest
 => => unpacking to docker.io/library/ticket-app:latest

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> docker run ticket-app
Ticket Cancelled!
Refund Amount: ₹500.0

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system>
```

Kubernetes Deployment Configuration File

pom.xml (ticket-cancellation-system) | deployment.yaml | CancellationServiceTest.java | Dockerfile | Ticket.java

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ticket-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: ticket
  template:
    metadata:
      labels:
        app: ticket
    spec:
      containers:
        - name: ticket-container
          image: ticket-app
```

Kubernetes Pod Status (Running/Completed)

kubectl apply -f k8s/deployment.yaml

kubectl get pods

```
Terminal    Local

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> kubectl apply -f k8s/deployment.yaml
deployment.apps/ticket-deployment created

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> kubectl get pods
NAME                                      READY   STATUS             RESTARTS        AGE
bill-payment-deployment-7648ff8df6-pgjw5  1/1     Running            2 (39m ago)     3d1h
book-tracker-v1-6bc6886d6c-vltbc          1/1     Running            4 (39m ago)     3d23h
enrollment-deployment-6947745c7b-6w7mn    1/1     Running            6 (39m ago)     7d18h
enrollment-deployment-6947745c7b-9z7dr    1/1     Running            6 (39m ago)     7d18h
mfa-app-5d879b7cc-n2xzw                   1/1     Running            12 (39m ago)    6d23h
route-optimizer-78699fdbf9-lmwt7          0/1     CrashLoopBackOff   66 (3m9s ago)   2d23h
ticket-deployment-7686d8fd9b-5pqzg        0/1     Completed          1 (5s ago)      7s

PS C:\Users\Sabarishwaran\IdeaProjects\ticket-cancellation-system> kubectl logs ticket-deployment-7686d8fd9b-5pqzg
Ticket Cancelled!
Refund Amount: ₹500.0
```